Predicting diabetes status with statistical models

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1. INTRODUCTION

This report develops predictive models for the binary outcome **`diabetes`** (0/1) using a large tabular dataset. The analysis proceeds in two stages. First, ***Exploratory Data Analysis (EDA)*** describes the dataset and class balance, and examines first-order relationships between predictors and the outcome using standard visuals: a bar chart for the outcome; row-normalised stacked bar charts for categorical variables; and, for numerical variables, overlayed class histograms together with boxplots to display distributional shifts and summary spread. These descriptive views motivate modelling choices without formal hypothesis tests. Second, ***baseline classifiers*** - *k*-Nearest Neighbours (*k*-NN), Decision Tree (DT), and Logistic Regression (LR) - are trained on a shared five-fold cross-validation split. Categorical predictors are encoded as factors; numerical predictors are standardised where appropriate for distance-based methods (*k*-NN). Within each method, the best version is selected by Type I error (FPR) on the fixed five folds (e.g., best k for *k*-NN; suitable depth/minsplit for DT; an LR that is not overly complex). The three best versions are then refit and compared on the full dataset, with true positive rate (TPR), FPR, ROC-AUC *(Receiver Operating Characteristic - Area Under the Curve)* reported.

2. EXPLORATORY DATA ANALYSIS

2.1 Description of dataset

The dataset contains 100,000 observations with a binary response diabetes (0 = non-diabetic, 1 = diabetic) and 8 different features (Table 1). The dataset is also clean with no missing values; categorical variables are consistently coded and analysed as factors, and numerical variables load with plausible ranges. The outcome is strongly imbalanced, with many more non-diabetic than diabetic cases (Figure 1).

feature	type	description	NA
gender	fct(3)	categorical	0
smoking_history	fct(5)	categorical	0
hypertension	fct(2)	categorical	0
heart_disease	fct(2)	categorical	0
age	num	numerical	0
bmi	num	numerical	0
HbA1c_level	int	numerical	0
blood_glucose_level	int	numerical	0
response	**type**	**value**	**count**
diabetes	fct(2)	0 (non-diabetic) 1 (diabetic)	91500 8500

Table 1. Dataset features

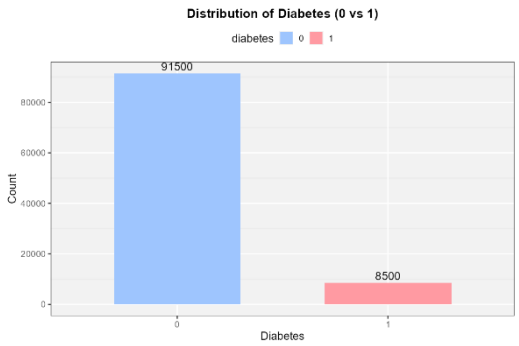

Fig.1. Distribution of response variable

2.2 Categorical variables

Categorical variables are examined using row-normalised stacked bar charts, read as the conditional proportions $P(\text{diabetes} = 1 \mid \text{level})$. To contextualise the stacked proportions, a faceted count plot is first shown to highlight uneven category sizes across variables (Figure 2).

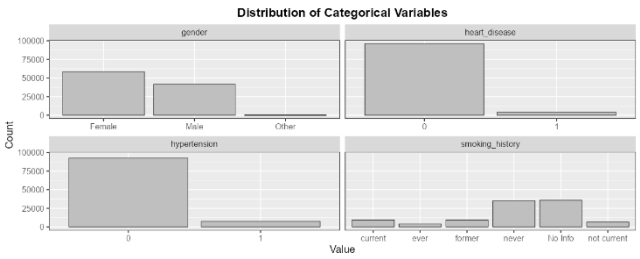

Fig.2. Distribution of categorical variables

Counts indicate that both levels of `hypertension` and `heart_disease` are well represented, while

`gender` is roughly balanced across its main levels; by contrast, several categories in `smoking_history` are relatively small.

For each variable, we report the percentage-point gaps between levels and interpret patterns as strong, moderate, or weak based on the magnitude and consistency of these gaps, with caution where category sizes are uneven. This descriptive approach avoids formal testing while making group-wise differences directly interpretable.

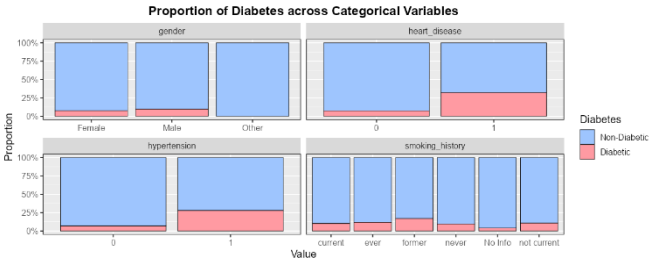

Fig.3. Relationship of categorical features to the response variable

`gender`. Proportions across gender levels are very similar, with no persistent directional shift. Any differences are small and not consistent, so the visual evidence is weak on its own.

`heart_disease`. A clear elevation is also observed when `heart_disease` = 1 relative to 0. The gap is sizeable and coherent, pointing to a strong association that complements hypertension rather than duplicating it.

`hypertension`. The diabetic share is consistently higher for `hypertension` = 1 than for 0, with a visibly large and stable gap across the chart. This pattern indicates a strong risk gradient and suggests meaningful standalone signal.

`smoking_history`. Differences between categories are small, and several levels are sparsely populated, which makes apparent gaps sensitive to re-weighting. Visual separation is weak and should be interpreted cautiously.

Summary. Among the categorical variables, `hypertension` and `heart_disease` display strong and consistent risk gradients and will be treated as informative. `gender` and `smoking_history` show weak standalone separation and will be retained primarily for completeness and potential stabilisation in modelling.

2.3 Numerical variables

Numerical variables are examined with overlaid class histograms (to see distributional shifts) and boxplots by outcome (to summarise location/spread and outliers). These visuals are read jointly: a large, consistent shift in medians and bulk mass with limited overlap indicates a strong signal; small shifts with substantial overlap indicate a weak standalone signal.

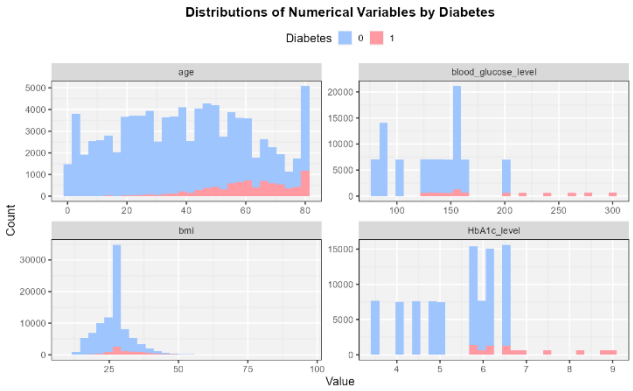

Fig.4. Distribution of numerical variables

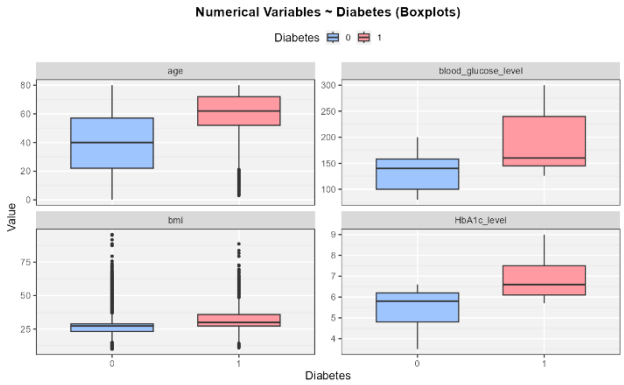

Fig.5. Relationship of numerical features to the response variable

`blood_glucose_level`. Diabetic cases concentrate at higher values, with a clearly higher median and wider IQR; the overlap with the non-diabetic group is limited. The histograms show a pronounced rightward shift. This is a strong predictor.

`HbA1c_level`. The pattern mirrors blood glucose: the diabetic group sits consistently higher with modest overlap. This is also a strong predictor and complements `blood_glucose_level`.

`age`. The diabetic group has a higher median age, but the distributions overlap substantially across the range. The evidence suggests a moderate effect that is useful in combination with stronger variables.

`bmi`. Distributions are right-skewed with many outliers; the diabetic median is only slightly higher and class overlap is pronounced. This provides a weak standalone signal.

Summary. Visual ranking of numerical signal is: `blood_glucose_level` ≈ `HbA1c_level` (strong) > `age` (moderate) > `bmi` (weak). All four variables

are retained for modelling, with numerical variables standardised for *k*-NN to avoid scale effects.

2.4 Conclusion

EDA indicates strong signals from `HbA1c_level` and `blood_glucose_level`, a moderate signal from `age`, and a weak signal from `bmi`; among categoricals, `hypertension` and `heart_disease` are informative, whereas `smoking_history and gender` are weak. However, all variables are retained (categoricals as factors; numericals standardised for *k*-NN) to avoid information loss and allow the models (DT/LR) to discover multivariate effects, and within-method tuning will use Type I error (FPR) on a shared five-fold split.

3. BUILDING MODELS (KNN, DT and LR)

A stratified ***10% subset*** is first drawn using `set.seed(611)`, sampling within each diabetes class (0/1) so the class ratio is preserved and all categorical levels (e.g., `smoking_history` = *"No Info"*, `gender` = *"Other"*) remain present. Three classifiers (k-Nearest Neighbours (KNN), Decision Tree (DT), and Logistic Regression (LR)) are then tuned on this subset using a single fixed ***5-fold cross-validation*** partition shared across methods. Categorical predictors are encoded as factors; for KNN, numerical predictors are standardised within each training fold and the same centring/scaling is applied to its validation fold to avoid information leakage.
The selection rule is the mean ***Type I error (FPR)*** across folds: choose k for KNN, `minsplit` for DT, and choose an LR variable set that balances fit with simplicity. A common decision threshold $\delta = 0.5$ is used throughout tuning. The resulting best version per method (best k, best `minsplit`, chosen LR) is then carried forward for evaluation on the *full dataset* using TPR, precision, ROC, and AUC, with FPR reported for context.

3.1 k-Nearest neighbours (k-NN)

Methodology. For an input vector $x \in \mathbb{R}^p$, define the neighbourhood $N_k(x)$ as the set of the k training observations with the smallest *Euclidean distances* to x:

$$d(x, x_i) = \sqrt{\sum_{m=1}^{p} (x_m - x_{im})^2}.$$

Let the binary class labels be $y_i \in \{0,1\}$. The KNN score is the neighbour average, and the decision rule is the usual majority vote:

$$\hat{G}(x) = \begin{cases} 1, & \hat{y}(x) > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

$$\hat{y}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i,$$

KNN is non-parametric; classification is determined by local neighbourhood composition, not by an explicit fitted coefficient vector.

Predictors used. Because distance is defined in a continuous space, KNN is fitted only on *numerical variables*: `age`, `bmi`, `HbA1c_level`, `blood_glucose_level`. Categorical variables are excluded from the KNN distance.

Standardisation. To ensure that all numerical variables contribute on a comparable scale, each variable is *z-standardised* within the training fold and the same transformation is applied to the corresponding test fold. For variable X_m,

$$z_{im} = \frac{x_{im} - \mu_m^{(\text{train})}}{\sigma_m^{(\text{train})}},$$

where $\mu_m^{(\text{train})}$ and $\sigma_m^{(\text{train})}$ are the mean and standard deviation computed using only the training observations in that fold. This prevents information leakage and removes scale dominance in the Euclidean metric.

Cross-validation and selection rule. A single 5-fold cross-validation partition is fixed and reused. Candidate k values are *odd integers* to reduce ties. For each k, 5-fold CV records the *mean Type I error (FPR)* across folds as the tuning criterion (definitions: $\text{FPR} = \frac{\text{FP}}{\text{FP+TN}}$; $\text{TPR} = \frac{\text{TP}}{\text{TP+FN}}$)

Result. The CV curve of mean decreases then plateaus for $k \approx 25$–35. The grid minimum is at $k = 33$ (FPR = 0.00120). At $k = 29$, the FPR is 0.00142 (absolute + 0.00022) while TPR is higher

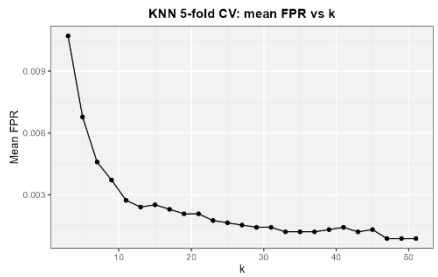

Fig.6. FPR performance of *k*-NN models

(0.588 vs 0.585) and accuracy similar. With Type I error as the primary criterion and differences on the plateau negligible, $\boldsymbol{k = 29}$ is adopted, near-minimal FPR with slightly better TPR, ACC and a smaller k.

k	FPR	TPR	ACC
3	0.010710	0.669412	0.962100
5	0.006776	0.647059	0.963800
7	0.004590	0.625882	0.964000
9	0.003716	0.605882	0.963100
11	0.002732	0.609412	0.964300
13	0.002404	0.604706	0.964200
15	0.002514	0.601176	0.963800
17	0.002295	0.603529	0.964200
19	0.002077	0.608235	0.964800
21	0.002077	0.600000	0.964100
23	0.001749	0.592941	0.963800
25	0.001639	0.594118	0.964000
27	0.001530	0.589412	0.963700

k	FPR	TPR	ACC
29	0.001421	0.588235	0.963700
31	0.001421	0.590588	0.963900
33	0.001202	0.584706	0.963600
35	0.001202	0.578824	0.963100
37	0.001202	0.580000	0.963200
39	0.001311	0.578824	0.963000
41	0.001421	0.574118	0.962500
43	0.001202	0.570588	0.962400
45	0.001311	0.569412	0.962200
47	0.000874	0.565882	0.962300
49	0.000874	0.564706	0.962200
51	0.000874	0.563529	0.962100

Fig.7. Summary FPR, TPR, ACC of of *k*-NN models

3.2 Decision Tree (DT)

Methodology. A classification tree is constructed by *recursive binary splitting* of the feature space. At each node, all candidate splits on all predictors are evaluated using the ***information*** criterion (entropy); the split that most reduces class impurity is selected. Splitting proceeds top-down until a *stopping rule* is met. Leaves predict the *majority class* in the node (default 0.5 threshold). Categorical predictors enter as factors; numerical predictors are used directly (no standardisation required for trees).

Tuning and selection. Using the same fixed 5-fold cross-validation partition as other methods, a grid of minsplit ∈ {10, 15, …, 100} is evaluated. For each value, the Type I error (FPR) on the held-out fold is recorded and then averaged across folds. The selection rule follows the coursework specification: retain the version with the smallest mean FPR; TPR and ACC are reported for context.

minsplit	FPR	TPR	ACC
10	0.017596	0.751765	0.962800
15	0.013552	0.740000	0.965500
20	0.010383	0.737647	0.968200
25	0.007869	0.730588	0.969900
30	0.006120	0.725882	0.971100
35	0.005137	0.721176	0.971600
40	0.005574	0.721176	0.971200
45	0.004372	0.717647	0.972000
50	0.003607	0.716471	0.972600
55	0.002951	0.709412	0.972600

minsplit	FPR	TPR	ACC
60	0.002623	0.708235	0.972800
65	0.002623	0.708235	0.972800
70	0.001858	0.709412	0.973600
75	0.001858	0.709412	0.973600
80	0.001858	0.709412	0.973600
85	0.001858	0.709412	0.973600
90	0.001530	0.703529	0.973400
95	0.001530	0.703529	0.973400
100	0.001530	0.703529	0.973400

Fig.8. Summary FPR, TPR, ACC of of DT models

Result. A 5-fold sweep over minsplit = 10…100 (`split = "information"`) shows a sharp fall in mean FPR up to ≈*60*, then a *clear plateau*. Concretely, increasing minsplit from 60 to 100 reduces FPR only from 0.00262 to 0.00153 (Δ ≈ 0.00109) while TPR declines from 0.708 to 0.704. By contrast, smaller settings (≈45–55) incur higher FPR (≈0.0044–0.0030) for only marginal TPR gains (≈0.7176–0.7094). Balancing low FPR within the plateau against unnecessary TPR loss and model complexity, **minsplit = 60 is selected**. The resulting tree remains compact and interpretable, with primary splits aligned to the strongest numer-ical signals identified in the EDA (`HbA1c_level, blood_glucose_level`).

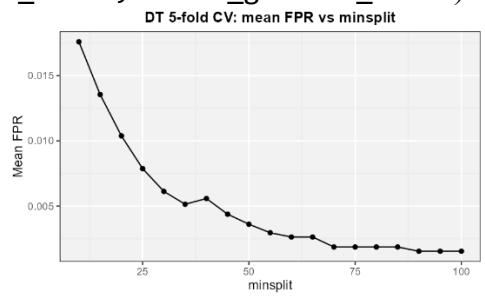

Fig.9. FPR performance of *k*-NN models

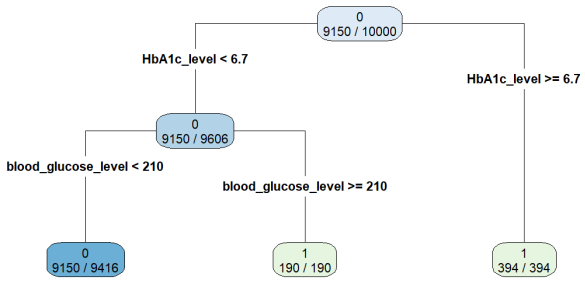

Fig.10. 60-minsplit decision tree

3.3 Logistic Regression (LR)

Methodology. Let $Y \in \{0,1\}$ indicate diabetes and let $p(x) = \Pr(Y = 1 \mid x)$. LR models the **log-odds** ("logit") as a linear score

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta^\top x, \quad \hat{y} = \mathbf{1}\{\hat{p}(x) \geq 0.5\}.$$

Coefficients are estimated by maximum likelihood. To balance fit and simplicity, a ***backward removal*** is applied using the p-values reported in the model summary: start from the full model and iteratively drop variables with the largest non-significant p-values, re-fitting after each drop. Models are compared using the same fixed 5-fold CV as KNN/DT; the selection rule is the mean Type I error (FPR) across folds (TPR, ACC for context). Decision threshold fixed at 0.5.

Backward regression.

- **M1 (Full):** all eight predictors
- **M2:** drop `smoking_history`
- **M3:** drop `smoking_history` and `gender`

Fitting on the subset, **M3** yields all retained coefficients significant; signs are clinically sensible (older `age`, higher `bmi`, higher `HbA1c` and `blood_glucose`, and indicators for `hypertension/heart_disease` increase the odds of diabetes). Although `bmi` appears relatively weak in the univariate EDA, it is *significant jointly* once other covariates are controlled, so it is kept.

Variables	Estimate	p-value
(Intercept)	-28.634805	< 2e-16
age	**0.053828**	**< 2e-16**
hypertension1	0.624438	2.94e-05
heart_disease1	0.633361	0.000659
bmi	**0.096419**	**< 2e-16**
HbA1c_level	**2.360893**	**< 2e-16**
blood_glucose_level	**0.036902**	**< 2e-16**

bold values indicate the most significant features ($\alpha = 0.05$)

Table 2. Description of the M3 logistic model

Cross-validation performance. The bar chart reports the *mean FPR* over folds for each model; *TPR* and *ACC* (below) are listed for context.

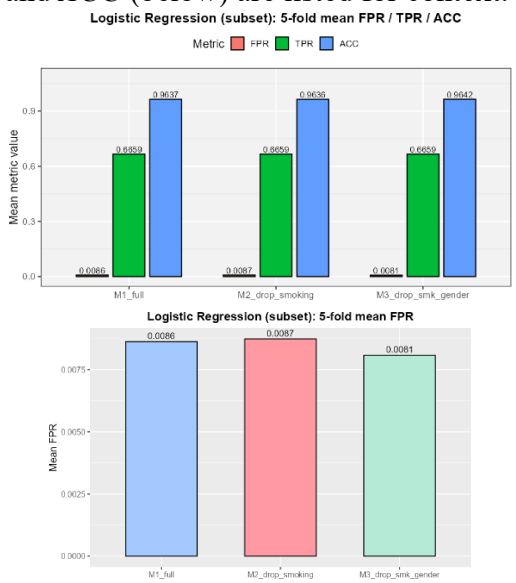

Fig.11. Bar charts of mean FPR, TPR, ACC for LR models

Results. M3 (drop `smoking_history` and `gender`) is chosen: it achieves the lowest mean FPR while keeping TPR unchanged and slightly improving ACC, and it is simpler than M1/M2. This specification is carried forward to ROC/AUC and later comparison with KNN and DT.

4. MODEL COMPARISON & FULL-DATA EVALUATION

4.1 Experimental setup

The retained versions from Part III, KNN ($k = 29$), Decision Tree (DT; *minsplit* = 60, split = "information"), and Logistic Regression (LR; M3: `age, hypertension, heart_disease, BMI, HbA1c_level, blood_glucose_level`), are evaluated on the full data (≈100,000). A single stratified 5-fold partition is fixed and reused across methods. For point metrics, the decision threshold is $\delta = 0.5$. Reported measures are Type I error (FPR), True Positive Rate (TPR), Precision, Accuracy (ACC), and from pooled out-of-fold scores, ROC curves with AUC.

4.2 Aggregate metrics

Point metrics (ACC, FPR, Precision, TPR).

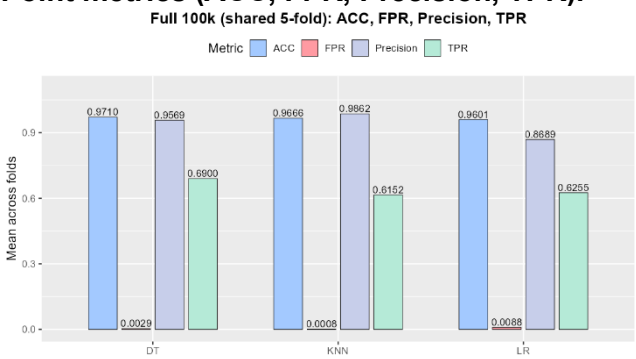

Fig.12. Mean metrics of full 100k (shared 5-fold) for the best three models

Figure 12 reports fold-means of ACC, FPR, Precision, and TPR for the three classifiers. Overall performance is high and tightly clustered. KNN achieves the lowest FPR (~0.0008) and highest precision (~0.9862), but with the lowest TPR (~0.6151). DT delivers the highest Accuracy (~0.9710) and highest TPR (~0.6900) while keeping FPR small (~0.0029). LR lies between KNN and DT on TPR (~0.6255) but has the highest FPR (~0.0088) and the lowest Accuracy (~0.9601). Absolute gaps are small at the reported threshold $\delta = 0.5$, though the trade-offs differ by method.

4.3 ROC curves and AUC

To summarise threshold-free discrimination, out-of-fold class-1 scores are pooled across the five test folds and **ROC curves** are drawn for each method (Figure 13). The corresponding **AUCs** are:

- **DT:** AUC = **0.9691**
- **LR:** AUC = **0.9606**
- **KNN:** AUC = **0.9592**

DT's ROC dominates slightly in the high-specificity region; LR tracks closely; KNN is effectively on par with LR at most operating points.

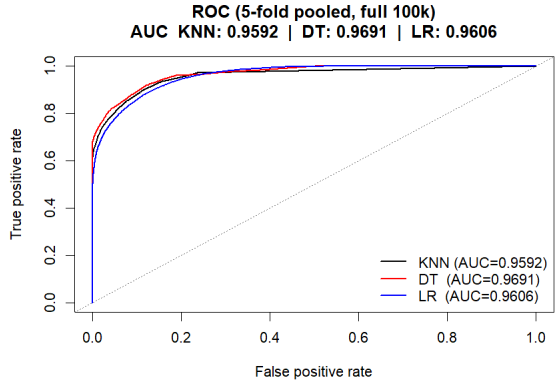

Fig.13. ROC curves and AUCs for three models

4.4 AUC significance (DeLong test)

Because the three ROC curves are computed on the same out-of-fold observations, pairwise differences in AUC are assessed using DeLong's test (a non-parametric test for two correlated ROC curves):

- **LR vs DT:** $Z = -12.072$, $p < 2.2\times10^{-16}$; 95% CI for AUC (LR) – AUC (DT): [−0.0098, −0.0071].

  ⟶ DT's AUC is *higher* than LR's (gap ≈ 0.0084).
- **LR vs KNN:** $Z = 1.5246$, $p = 0.1274$; 95% CI for AUC (LR) – AUC (KNN): [−0.0004, 0.0034].

  ⟶ LR and KNN show *no statistically significant* difference in AUC.

Interpretation. The three models are all strong; DT shows a small but statistically reliable AUC edge over LR on this full-data evaluation, whereas LR and KNN are statistically indistinguishable by AUC. Precision/ TPR/ ACC at δ = 0.5 are closely matched, so any preference among LR and KNN is better grounded in considerations of interpretability and deployment cost rather than raw fit.

5.1. COMPARATIVE APPRAISAL & CONCLUSION

5.1 k-Nearest Neighbours (KNN, k = 29)

Pros. Non-parametric; competitive discrimination on full 100k (AUC ≈ your pooled result), with balanced TPR/ACC once δ = 0.5 is fixed; trivial to explain operationally ("vote of neighbours").

Cons. Works only on numeric space (categoricals excluded from distance); requires per-fold standardisation; prediction cost scales with training size; difficult to attribute effects to predictors; performance sensitive to local density and class imbalance.

5.2 Decision Tree (DT, minsplit = 60)

Pros. Interpretability via rules; fast inference; low FPR at the default threshold in this sample; natural handling of mixed types.

Cons. High variance/ instability to sampling and small perturbations; step-wise probabilities with weak calibration; gains over LR are narrow - your pooled ROC shows only a small AUC edge.

5.3 Logistic Regression (LR, M3)

Pros. Parsimony and stability (six predictors after backward removal); well-calibrated probabilities; coefficients align with EDA (age, BMI, HbA1c, glucose, hypertension, heart disease all positive and significant), enabling transparent policy control of FPR via threshold; cheapest to deploy and monitor.

Cons. Linear log-odds may miss interactions/ nonlinearities unless engineered; assumes additivity on the logit scale.

5.4. Conclusion

Synthesis across operating metrics. On the shared 5-fold over the full data, LR's *FPR/ TPR/ACC/Precision* are competitive with KNN and close to DT; DT's small AUC lift does not translate into a dominant advantage at δ = 0.5 and comes with higher instability and poorer calibration. KNN matches discrimination but lacks interpretability and incurs heavy inference.

Recommendation. Considering discrimination (ROC/AUC), operating-point control (FPR at δ = 0.5), calibration, interpretability, and deployment cost, the most defensible choice for this dataset is **Logistic Regression (M3)** as the final classifier. DT's slight AUC lead is statistically significant yet **practically modest**; LR offers the best balance for transparent reporting and downstream use.